• STEERING—WANDERING/STIFF STEERING OR POOR RETURNABILITY AFTER TURNS—4X4 VEHICLES ONLY • SUSPENSION—WANDERING/STIFF STEERING OR POOR RETURNABILITY AFTER TURNS—4X4 VEHICLES ONLY	Article No. 01-3-2

FORD: 1999-2001 SUPER DUTY F SERIES
2000-2001 EXCURSION

ISSUE

Some vehicles may exhibit stiff steering, wandering, or poor returnability after turns. This may be caused by alignment, steering gear meshload, or a lower knuckle ball joint that becomes sticky and degrades steering returnability.

ACTION

Check alignment and check steering gear meshload or install greaseable ball joint and lube every 8000 kilometers (5000 miles) or at every oil change (whichever comes first). Refer to the following Service Procedure for details.

SERVICE PROCEDURE

Verify stiff steering, wandering, or poor returnability after turns. Proceed as follows:

1. Check alignment and correct if not to specification.

NOTE
USE 0.75 DEGREE OR LESS CASTER SLUGS TO CHANGE THE CROSS-CASTER. IF THE CASTER REQUIRES MORE THAN A 0.75 DEGREE CHANGE YOU MUST USE CASTER WEDGES THAT ARE INSERTED BETWEEN THE AXLE AND LEAF SPRINGS. USE IDENTICAL CASTER WEDGES ON BOTH SIDES TO CHANGE CASTER.

   a. If alignment corrects the condition, return vehicle to customer.

   b. If alignment does not resolve condition, continue to Step 2 (steering knuckle torque to turn test).

2. Perform the following steering knuckle torque to turn test:

   a. Place vehicle on alignment rack with front wheels on low friction plates. Wheels must be pointing straight forward.

   b. Remove tie rod from steering knuckle. Install a nut and bolt into the steering arm hole and tighten.

NOTE
WHEELS MUST BE LOADED AND ON PROPERLY OPERATING TURN PLATES TO PROPERLY CHECK TURNING TORQUE.

   c. Use a click-style torque wrench and position it on the bolt head so that the handle is pointing toward the front of the vehicle.

   d. Set the wrench at 135 N•m (100 lb-ft).

   e. Pull on the torque wrench so that you are turning the front wheel outboard.

      (1) If the knuckles turn without the wrench clicking, then ball joint turning torque is acceptable. Return vehicle to customer.

      (2) If the wrench clicks before the wheel breaks free and begins to turn, then the torque to turn the knuckle is too high. Continue by replacing ball joints (Step 3).

   f. Perform this procedure on both front wheels.

3. If the torque wrench clicks before the wheel breaks free and begins to turn, then replace the lower ball joint with the greaseable Lower Ball Joint (1C3Z-3050-AB) and replace the upper with the current replacement part (F6TZ-3049-AA). Refer to the appropriate model/year Workshop Manual for removal and installation procedures.

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Follow this procedure to torque the ball joints after knuckle installation.

a. Torque lower ball socket nut to 47 N•m (35 lb-ft).

b. Torque upper ball socket nut to 95 N•m (70 lb-ft), then advance nut to the next slot to line up with hole and install cotter pin.

c. Retorque lower ball socket nut to 190/217 N•m (140/160 lb-ft).

NOTE

THE NEW GREASEABLE LOWER BALL JOINT MUST BE LUBRICATED AFTER INSTALLATION.

SERVICE SCHEDULE FOR GREASEABLE LOWER BALL JOINT

If a greaseable lower ball joint is installed, it must be serviced per the below schedule:

- At every oil change or at 8000 kilometer (5000 mile) intervals (whichever comes first)

The recommended lubricants are:

- Premium Long Life Grease (XG-1-C, or equivalent meeting Ford Specification ESA-M1C75-B)
- Hi-Temp Wheel Bearing Grease (E8TZ-19590-A, or equivalent meeting Ford Specification ESA-M1C198-A)

4. Measure steering gear meshload. Reference the appropriate Workshop Manual for the procedure for measuring meshload.

   a. If it is less than 0.45 N•m (4 lb-in), reset to 0.56-0.68 N•m (5-6 lb-in).

NOTE

THE MESHLOAD MEASUREMENT IS THE AMOUNT OF THE TORQUE INCREASE THAT THE TORQUE WRENCH SHOWS AS THE STEERING WHEEL IS ROTATED THROUGH THE CENTER OF ITS TRAVEL (I.E., IF THE STEERING WHEEL TORQUE TO ROTATE IS MEASURED AT 0.45 N•m (4 LB-IN) ON THE ENDS, AND THEN IT MEASURES 1.13 N•m (10 LB-IN) AS IT IS ROTATED THROUGH THE CENTER, MESHLOAD = 0.68 N•m (6 LB-IN). MESHLOAD = CENTER - END (1.13 N•m - 0.45 N•m = 0.68 N•m OR 10 LB-IN - 4 LB-IN = 6 LB-IN OF MESHLOAD).

   b. Once meshload is OK, procedure is complete. Return vehicle to customer.

PART NUMBER	PART NAME
1C3Z-3050-AB	Ball Joint - Lower (Greaseable Type)
F6TZ-3049-AA	Ball Joint - Upper
XG-1-C	Premium Long Life Grease
E8TZ-19590-A	Hi-Temp Wheel Bearing Grease

OTHER APPLICABLE ARTICLES: NONE

WARRANTY STATUS: Eligible Under The Provisions Of Bumper To Bumper Warranty Coverage

OPERATION	DESCRIPTION	TIME
010302AB	Additional Time To Perform Turning Torque Test (Includes Both Front Wheels)	0.5 Hr.
010302AC	Additional Time To Replace Upper And Lower Ball Joints	2.6 Hrs.
010302AD	Additional Time To Measure And Adjust Meshload As Required	0.4 Hr.
010302A	Verify Stiff Steering Complaint (Includes Time To Check Alignment And Correct If Required)	2.2 Hrs.

DEALER CODING

BASIC PART NO.	CONDITION CODE
3050	42

OASIS CODES: 303000